

CHAPTER 2

The Cell: The Fundamental Unit in Systems Biology

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INTRODUCTION

The cell is the fundamental unit in the systems biology/toxicology paradigm. Exposure to environmental contaminants may result in tissue, organ, or systemic toxicity in part through altered cellular homeostasis. Specifically, global responses, whether protective or deleterious, originate from the individual responses of cells. This view of the importance of the cell is appropriate in the context of systems biology which seeks to uncover cellular relationships and networks in order to understand the response of the total system. Therefore to predict the systemic effects of contaminant exposure, the molecular mechanisms of cellular homeostasis must be understood, for cellular responses inform the methodologies employed in systems biology. The cell provides a quantifiable response (changes in gene or protein expression) that through systems-level science is incorporated into an integrative framework (adverse cellular or tissue response) to link causative factors such as toxicant exposure with a biological outcome (disease) ([Figure 1](#)).

A systems biological perspective utilizes cellular responses in the form of alterations in the cellular genetic profiles (**genome**), global messenger RNA ribonucleic acid (mRNA) expression profiles (**transcriptome**), global protein expression profiles (**proteome**), total metabolite levels (**metabolome**), and the epigenetic alteration signature (**epigenome**). Therefore, this chapter will focus primarily on the process by which the genome is transcribed into the transcriptome, and subsequently translated into the proteome, with emphasis on intracellular signal transduction and epigenetics, which initiate and regulate transcription and translation, respectively.

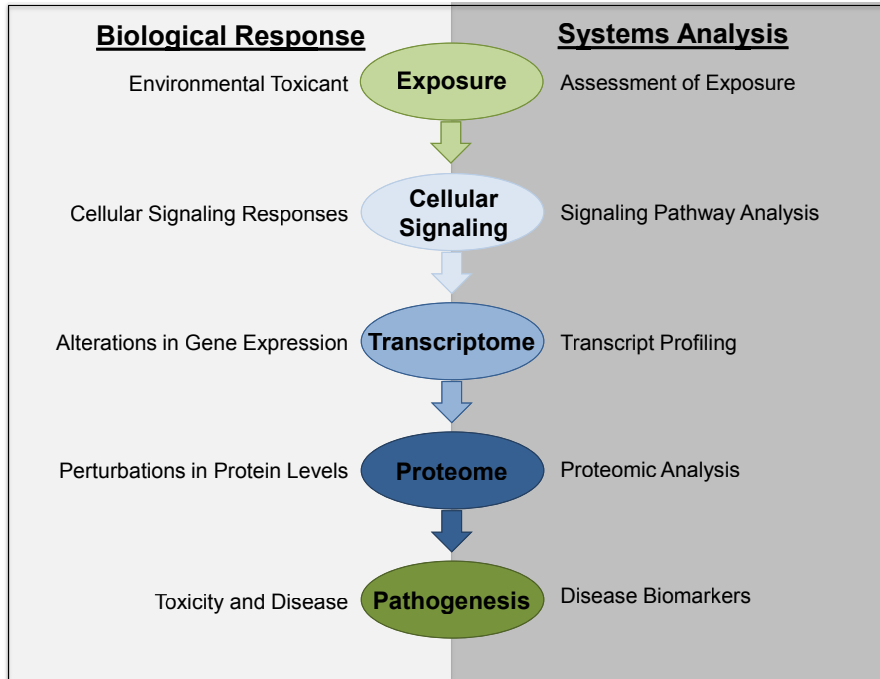


Figure 1 Flow of biological response information in a systems biology framework.

THE CELL: STRUCTURE AND ORGANELLES

Cells are the primary biological unit of living organisms, with the ability to faithfully pass along to daughter cells the parental genetic information. As this textbook focuses on the effects of environmental contaminants on human health, this chapter focuses on eukaryotic cells. Eukaryotic cells are larger than simple prokaryotic cells, and are more complex, containing membrane-bound subcellular structures called **organelles**. The genetic code of eukaryotes is enclosed in a membrane-bound organelle called the **nucleus**; this is a major feature distinguishing eukaryotic cells from prokaryotes. Cellular organelles are surrounded by the cytoplasm, which in turn is contained by an outer cell membrane, or **plasma membrane**, which separates the inner cell from the environment (Figure 2).

Macromolecules

The structural and functional components of the cell are polymeric macromolecules, assembled from carbon-based monomers. Carbon's unique property of being able to bond with itself and other atoms make it the core element from which a vast number of molecules of diverse structure and function are formed. The increasing complexity of structure and function arises from the addition of oxygen, nitrogen, sulfur,

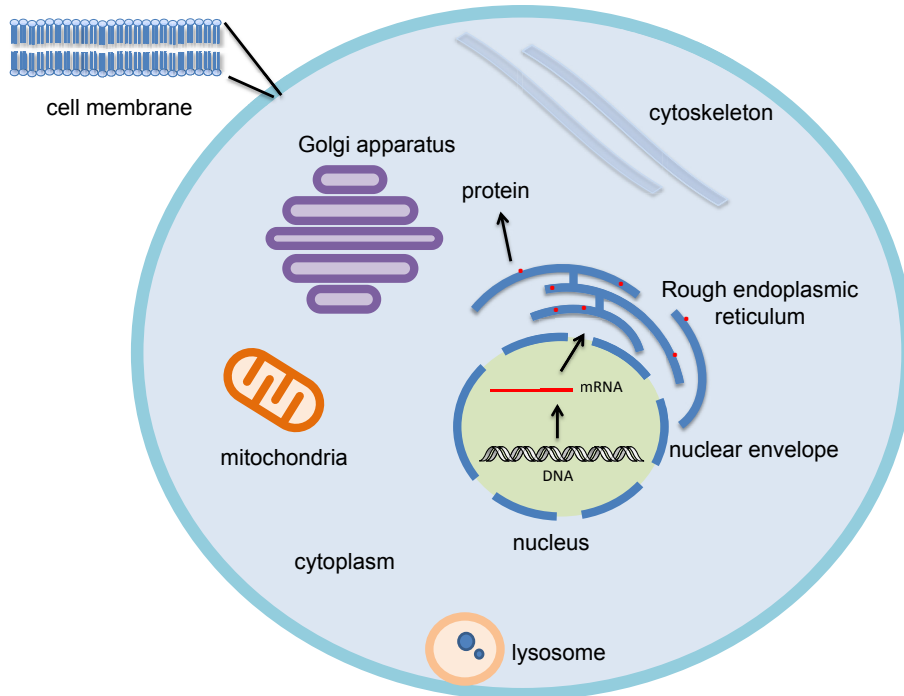


Figure 2 *The eukaryotic cell.*

hydrogen, and phosphorous atoms. The cell is mainly composed of four distinct classes of macromolecules: **carbohydrates**, **lipids**, **nucleic acids**, and **proteins**.

Carbohydrates

Carbohydrates, or sugars, are key sources of energy. They are generally categorized into three classes: monosaccharides, disaccharides, and polysaccharides. Simple sugars, or monosaccharides, are categorized based on the number of carbon atoms they contain. Examples are glucose and ribose. Disaccharides are composed of two linked monosaccharides, such as sucrose, while polysaccharides are made up of large numbers of monosaccharides. Glycogen is an energy storing polysaccharide composed of glucose monomers.

Lipids

Lipids are a structurally diverse class of biomolecules whose common characteristic is low solubility in water. This class includes oils and fats and is loosely grouped by structure. For example, fatty acids and phospholipids are members of those lipids with an open chain structure, having a hydrophobic nonpolar “tail” and a charged polar “head” component. Many biological membranes are composed of this class of lipids, as will be

discussed in the section on the cell membrane. The second major class is ringed lipids, known collectively as steroids, of which cholesterol is a member. Steroids serve important roles in intercellular signaling.

Nucleic Acids

Nucleic acids carry the genetic “code” of the cell. Nucleic acids are polymers of nucleotides: a sugar (deoxyribose or ribose) ring, phosphate groups, and a purine or pyrimidine base. **Deoxyribonucleic acid (DNA)** encodes the genetic information of the cell. DNA is composed of a deoxyribose ring and one of four bases: adenine, guanine, cytosine, and thymine. These four bases comprise the genetic code. **Ribonucleic acid (RNA)** is transcribed from DNA, the “print-out” of the genetic code. It is RNA that directs protein synthesis.

Proteins

Proteins, or polypeptides, are a chain of amino acids whose sequence is dictated by the RNA transcript. Proteins are the workhorses of the cell, carrying out a vast array of cellular functions from structure to enzymatic reactions. Polypeptides are assembled at ribosomes on the **rough endoplasmic reticulum (rER)** and are processed and folded into primary, secondary, and tertiary structures.

Organelles

Plasma (Cell) Membrane

The plasma membrane forms the exterior of the cell, separating the cytosol and nucleus from the environment (Figure 2). The cell membrane serves to protect the cell and regulates transport of molecules in and out of the cell. The plasma membrane is primarily composed of a lipid bilayer and transport proteins.

Cytoplasm

The cytoplasm is the cytosol and the organelles of the cell. The cytoplasm is bound by the plasma membrane. Cytosol is the aqueous component in which the organelles inhabit the cell.

Cytoskeleton

The cytoskeleton is a dynamic network of protein filaments that defines the structure and motility of the cell (Figure 2). It prevents deformation of the cell, while allowing for cell-mediated changes in structure including contraction and cell migration. The cytoskeleton is involved in cellular signaling and mediates chromosome segregation and cytokinesis in mitosis. It affects intracellular movement of organelles and vesicles and serves as a scaffold for cellular contents, interacting heavily with the plasma membrane. The eukaryotic cytoskeleton is composed primarily of three types of filaments:

microfilaments, intermediate filaments, and microtubules. Microfilaments are composed of linear actin polymers. Intermediate filaments are composed of a variety of different proteins. Microtubules are hollow structures composed of tubulin.

Nucleus

The nucleus is the genetic command and control center of the cell. Each cell contains the heritable information needed to direct cellular activity. This “master plan” is the same for every cell in an organism, though different cell types have different functions and therefore each type of cell uses different parts of the blueprint. The entire genome of the cell (with the exception of mitochondrial DNA) is packaged in the nucleus as **chromatin** in order to accommodate the large size of the genome. Chromatin is separated from cytoplasm by a double-layered **nuclear envelope** (Figure 2), which also regulates transport of molecules in and out of the nucleus, such as transcription factors and transcribed RNA.

Mitochondrion

The energy demands of the cell are met intracellularly through the generation of adenosine triphosphate (ATP). The mitochondrion is a double-membrane cytoplasmic organelle in which ATP is generated (Figure 2) through a process known as cellular respiration by oxidative phosphorylation in the electron transport chain. The mitochondrion is composed of two membranes: an outer membrane and an inner membrane. These two membranes encapsulate the intermembrane space, and the interior space enclosed by the inner membrane is the matrix. The matrix contains the enzymes that convert sugars and fats into energy—the mitochondrion is the “power plant” of the cell, and is the only organelle other than the nucleus to contain DNA. The mitochondrion also plays major roles in cell death.

Endoplasmic Reticulum (ER) and Golgi Apparatus

The endoplasmic reticulum is a membranous organelle that is the site of protein synthesis. The ER is composed of flattened, interconnected membrane-bound sacs called cisternae. There are two types of ER: smooth ER which carries out lipid metabolism, and rough ER, which is studded with ribosomes, the site of protein synthesis. The rough ER membranes are continuous with the nuclear envelope, as necessary for mRNA translation by ribosomes (Figure 2). The Golgi apparatus processes proteins, especially for secretion. The Golgi apparatus is composed of several flattened cisternae. The *cis* face is oriented toward the ER and receives vesicle-bound proteins from the ER. The *trans* face is oriented away from the ER and releases modified proteins in secretory vesicles destined for export from the cell (Figure 2).

Lysosomes and Peroxisomes

Lysosomes and peroxisomes are membrane-bound vesicles that break down macromolecules. Lysosomes contain a multitude of hydrolytic enzymes, serving to clean up

intracellular debris. Peroxisomes primarily break down fatty acids and are involved in energy metabolism (Figure 2).

CELLULAR SIGNALING: PLASMA MEMBRANE AND SIGNAL TRANSDUCTION

Cellular homeostasis is maintained by timely molecular responses to the demands of energy status or protection from toxic agents. How are these cellular signals relayed or transduced? Extracellular stimuli must first encounter and cross the plasma membrane in order to induce a cellular response. Examples of this response may be the upregulation of genes involved in growth in response to an extracellular growth factor, or an apoptotic response in the mitochondria in response to an environmental toxicant.

The Plasma Membrane

The plasma membrane serves several important functions. In conjunction with the cytoskeleton, the plasma membrane protects the interior of the cell from possibly harmful exogenous stimuli by presenting a selective, physiochemical barrier to the extracellular environment. In addition to protecting the cell from unwanted stimuli, the plasma membrane regulates transport of molecules entering and exiting the cell. Some molecules may pass through the lipid bilayer through simple diffusion while others need to be transported by specialized membrane-bound receptors. Through these receptors the plasma membrane relays signals that are passed through downstream signaling pathways. The plasma membrane therefore exists as an interface between extracellular signals or stimuli, and signal transduction systems in the cytoplasm, relaying signals from the extracellular environment to the nucleus or organelles.

Structure

The plasma membrane is composed of a phospholipid bilayer. The hydrophobic “head” of phospholipids orient towards the aqueous environment of the extracellular and cytoplasmic regions, while the hydrophobic hydrocarbon “tails” orient themselves inward toward each other, forming a semipermeable lipid bilayer (Figure 2). This hydrophobic barrier ensures the extracellular environment remains separate from the inner cytoplasm. Lipids account for roughly half of the membrane, the other half being composed of proteins.

Transport

Endocytosis and exocytosis are simple transport mechanisms. In **endocytosis**, the plasma membrane encircles material to be taken into the cell. **Exocytosis** is similar in nature, but employed to release material from the cell. There are two categories of transport, passive and active. **Simple diffusion** is a type of passive transport where the molecule

diffuses from a high concentration to a low concentration through the plasma membrane. The ability of a solute to cross the lipid bilayer depends upon the size and lipid solubility of the solute. An example of simple diffusion is the movement of oxygen across the plasma membrane, called osmosis. Another type of passive transport involves hydrophilic or charged particles crossing through a protein channel embedded in the plasma membrane. Facilitated diffusion is another passive transport mechanism where larger molecules are “carried” by single transport proteins. While passive transport is typified by the lack of an energy requirement, **active transport** requires energy in the form of ATP. Energy is necessary because active transport is against an electrochemical gradient, transporting from low to higher gradients. Active transport employs membrane bound proteins that “pump” solutes out of the cell.

Membrane-Bound Receptors

Molecules may be transported into the cytoplasm, but some activate cellular signaling pathways by binding to receptor proteins, which serve as binding sites for ligands. There are several classes of receptors representing different classes of ligands, for example G-protein coupled receptors (GPCR) and receptor tyrosine kinase (RTK) (Figure 3) mediate the binding of a variety of ligands such as xenobiotics and growth factors.

Intracellular Signaling

The cell responds to external stimuli in many different ways. Some responses, such as the binding of a secreted growth factor, are homeostatic responses. Toxic responses are those such as a toxicant entering transport channels into the cytoplasm. Each type of response is inherently membranous; however, the control center of the cell, which will mount a defense, is the nucleus. How do the membrane and cytoplasmic organelles relay messages to the nucleus? Inter- and intracellular signaling is commonly referred to as **signal transduction**, the process by which cells communicate with the extracellular environment, other cells, and direct signals from one part of the cell to the other.

Intracellular signaling pathways are composed of various signaling mediators that can be activated or repressed depending upon the intent of the signal. Many signaling factors are protein kinases, enzymes that catalyze the phosphorylation of other proteins. Many signaling pathways are activated at the cell membrane where cell surface receptors bind intra- or extracellular ligands. This event activates subsequent “downstream” factors such as kinases, which are at the top of a multitier signaling cascade.

Phosphorylation by kinases activates several signaling factors, which in turn pass along the intended signal, ultimately leading to the activation or repression of transcription factors that regulate the expression of target genes. Cellular signaling pathways also regulate intracellular stimuli. The cell’s determination to undergo mitosis and divide, or increase the strength of a stress response, is realized through cellular signaling pathways. How does the cell “know” how to respond to one type of stimuli versus another? This

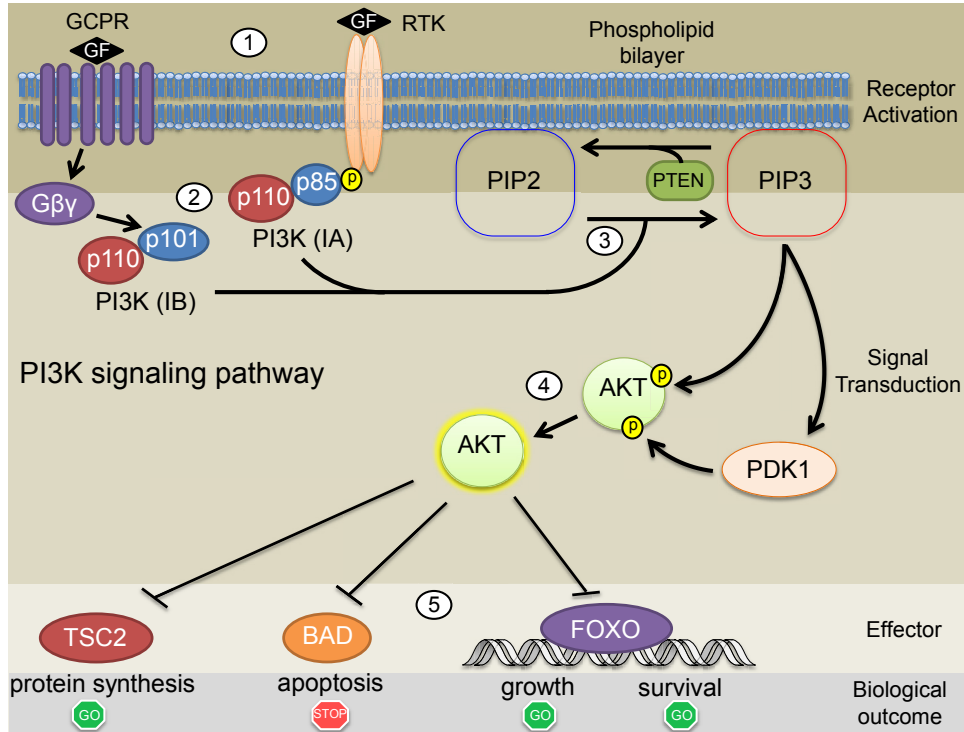


Figure 3 The PI3K signal transduction pathway. Ligand-activated cell membrane receptors RTK and GPCR (1) activate PI3KIA (p110/p85) and PI3KIB (p110/p101), respectively (2). Both forms catalyze the phosphorylation of PIP₂ to PIP₃ (3). PIP₃ recruits and activates signaling kinases AKT and PDK1 by interaction of PH domains (4). Activated AKT then regulates factors that control essential cellular homeostatic activities through transcription or factor stability or sequestration (5).

is resolved by the variety of signaling pathways in the cell. Signaling pathways may use the same class of signaling factors, whether membrane-bound systems or cytoplasmic kinases, but different pathways use several different types of signaling factors. Toxicants may alter the activity of cellular signaling pathways as a mechanism of toxicity or disease: toxicants may block ligand binding of receptors, alter the function of kinases, or regulate expression levels of pathway transcription factors. Thus, extracellular or intracellular signals are relayed through a linear cascade of enzyme interactions, resulting in the desired biological response.

The Phosphatidylinositol 3-Kinase Signaling Pathway

A classic example of a signal transduction pathway is the phosphatidylinositol 3-kinase (PI3K) signaling pathway. The PI3K pathway is a multitier signaling cascade that regulates cellular metabolism, growth, proliferation, differentiation, apoptosis, survival, and stress response (Figure 3). PI3K is a heterodimer composed of a catalytic and a regulatory

subunit. PI3Ks activated by RTKs are composed of the p110 catalytic subunit and the p85 regulatory subunit (class 1A). Growth factors activate cell surface RTKs, and the p85 subunit binds to phosphotyrosine residues on the activated RTKs [1]. Class 1B PI3Ks are composed of the p110 subunit and the regulatory p101 subunit, and are activated by interaction of p101 with the G-protein subunit $G\beta\gamma$, downstream of activated membrane-bound GPCRs [2].

Activated PI3K is recruited to the cell membrane where it catalyzes 3' hydroxyl group phosphorylation of the lipid phosphatidylinositol 4,5 bisphosphate (PIP_2) to phosphatidylinositol 3,4,5 triphosphate (PIP_3). PIP_3 recruits and activates pleckstrin homology (PH) domain-containing signaling kinases, such as phosphoinositide-dependent protein kinase 1 (PDK1) and protein kinase B (AKT). AKT, further activated by PDK1, phosphorylates forkhead family of transcription factor (FOXO), thus promoting cell proliferation and survival through inhibition of FOXO-targeted transcriptional events. AKT phosphorylates and inhibits tuberous sclerosis complex 2 and proapoptotic protein BCL2-antagonist of cell death (BAD), promoting protein synthesis and blocking apoptosis.

The PI3K pathway is negatively regulated by the dephosphorylation of PIP_3 to PIP_2 by the phosphatase and tensin homolog deleted on chromosome 10 (PTEN) phosphatase, thus inhibiting the activation of AKT and blocking downstream processes. Toxicants may disrupt signaling pathways at several key points, thus altering downstream biological processes; components of the PI3K pathway such as PTEN are frequently mutated in cancer, underlying the importance of signaling pathways in cellular homeostasis and environmental contaminant-induced toxicity and disease.

CELLULAR HOMEOSTASIS (I): ENERGY METABOLISM

As we require energy to breathe, think, and move, so do the cells that work in concert to carry out these functions. All of the cellular processes covered in this chapter, from cellular signaling to gene transcription, are the sum of millions of chemical reactions that require energy. The chemical reactions of the cell that are required to maintain homeostasis is known collectively as **metabolism**. Metabolic reactions that break down macromolecules from the food into simple molecules that serve as energy sources are **catabolic** reactions; those reactions that utilize this energy to assemble macromolecules from monomers are **anabolic** reactions.

ATP is the basic unit of energy “currency” used by cells. Cells extract energy by breaking down the chemical bonds of ATP, which is a nucleotide comprising a ribose ring, an adenine base, and three phosphate groups. The removal of an inorganic phosphate group (P_i) by hydrolysis releases a substantial amount of free energy ($ATP \rightarrow ADP + P_i + \text{Energy}$), while the addition of a phosphate group requires energy ($ADP + P_i + \text{Energy} \rightarrow ATP$) [3]. The cell may “withdraw” energy by removal of a

phosphate group, or “deposit” energy by the addition of a phosphate group. ATP is utilized in many reactions. In cellular signaling, kinases utilize the free energy and phosphate groups from ATP to phosphorylate target molecules. Energy from ATP hydrolysis is used to attach amino acids to transfer RNA (tRNA) in protein synthesis.

Cells catabolize macromolecules from food sources to produce energy in a process known as **cellular respiration** whereby oxygen serves as an electron acceptor. This is a redox (reduction/oxidation) reaction—reactions involving the loss or gain of electrons. A loss of electrons is an **oxidation** reaction while a gain of electrons is a **reduction** reaction. A loss of an electron usually involves the loss of an entire hydrogen atom; thus, a highly reduced, hydrogen-rich molecule is high in energy while a highly oxidized, hydrogen-depleted molecule is low in energy. For example, glucose ($C_6H_{12}O_6$) is a hydrogen-rich, highly reduced molecule that is oxidized for energy production. Cellular respiration is an oxidative process whereby an electron donor is oxidized and oxygen is reduced to produce carbon dioxide, water, and energy [3]. Before being transferred to oxygen however, the electrons are briefly transferred to nonprotein coenzyme carriers such as nicotinamide adenine dinucleotide (NAD^+) and flavin adenine dinucleotide ($FADH_2$). These carriers transfer high-energy electrons to the **electron transport chain**, which propels the synthesis of ATP. This reaction takes place in the inner membrane of mitochondria and is known as **oxidative phosphorylation**. The electrons necessary to drive the electron transport chain come from sugars and fatty acids through **glycolysis**, **fatty acid oxidation**, and the **citric acid cycle** (Figure 4).

During **glycolysis**, one molecule of glucose is converted into two molecules of pyruvate, with a net energy production of two molecules of ATP. Glycolysis occurs in the cytosol and consists of 10 steps, each catalyzed by a different enzyme. Initially, two molecules of ATP are utilized to convert the six-carbon glucose molecule into duplicate three-carbon sugars. These intermediate sugars are oxidized, forming two identical high-energy molecules. At this point, NAD^+ is reduced to NADH by accepting the electrons from the oxidation of the intermediate molecules. The two NADH coenzymes are employed later in the electron transport chain during oxidative phosphorylation. In the latter steps, high-energy phosphate groups are generated on each intermediate molecule, and are transferred to ADP, producing ATP. Each intermediate produces one NADH, and two ATP molecules (Figure 4).

Energy can be stored in the form of glycogen and fatty acids. Glycogen is a glucose polysaccharide that can be broken down to undergo glycolysis. Pyruvate from glycolysis is transported to the mitochondria where it undergoes pyruvate decarboxylation to produce acetyl coenzyme A (acetyl-CoA), NADH, and CO_2 . Fatty acids are a much more efficient energy storage unit, and produce more energy under oxidation than the same amount of glycogen. Low blood glucose levels cause the hydrolysis of triacylglycerols and the release of fatty acids from lipid droplets. Fatty acids are released into the bloodstream and transported to the mitochondria of cells. Fatty acid oxidation in the

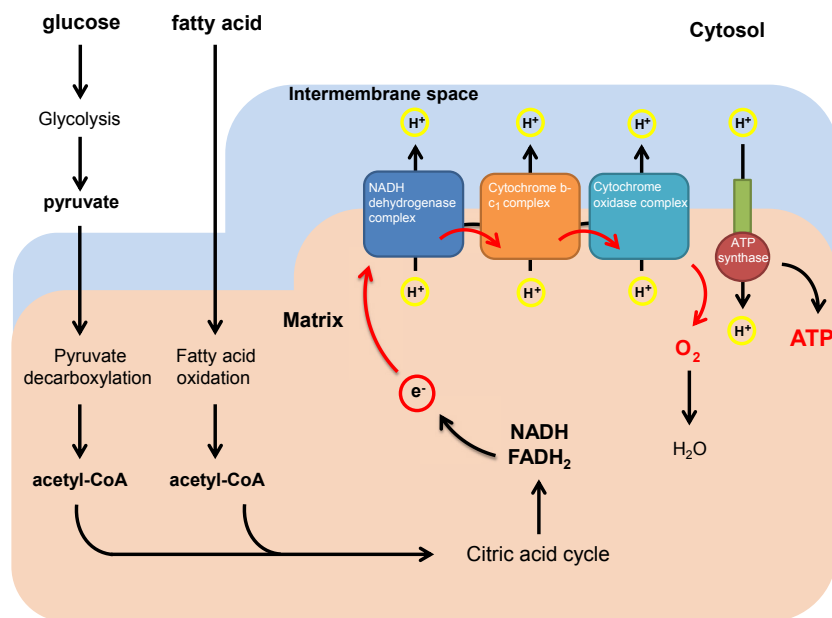


Figure 4 Cellular metabolism.

mitochondria produces acetyl-CoA, NADH, and FADH₂ in the fatty acid oxidation cycle. In the mitochondria, one molecule of acetyl-CoA is oxidized in the **citric acid cycle**, or **Kreb's cycle** (Figure 4), producing three NADH, one FADH₂, and one guanosine triphosphate (GTP) molecule(s). Electrons from NADH are used in the electron transport chain and eventually accepted by oxygen to generate water. Conversion of pyruvate and fatty acids into acetyl-CoA and the citric acid cycle occurs in the matrix, while the high-energy electrons from these processes are transferred in the form of NADH and FADH₂ to the inner membrane and the electron transport chain [4].

The electron transport chain in the inner membrane of the mitochondria catalyzes the oxidation of electrons carriers NADH and FADH₂, and this generates a hydrogen ion gradient that is utilized to produce ATP (Figure 4). The energy-producing capability of mitochondrial oxidative phosphorylation is much higher than glycolysis alone. NADH is oxidized to NAD⁺, releasing two high energy electrons that are passed along the electron transport chain carriers, which are metal ions bound by proteins forming large transmembrane complexes. There are three major complexes, the NADH dehydrogenase complex, the cytochrome b-c₁ complex, and the cytochrome oxidase complex. Each complex has greater affinity for electrons than the previous. After passing through these complexes, the electron is finally transferred to oxygen. Transfer of electrons from complex to complex drives the pumping of protons (H⁺) from the matrix, through the inner mitochondrial membrane, and into the intermembrane space. This generates a pH gradient (higher pH in the matrix) and a voltage gradient (negative in the matrix). The

pH gradient and voltage gradient propel H^+ through the membrane-bound ATP synthase back into the matrix; this is called an electrochemical proton gradient. ATP synthase harnesses this electrochemical gradient to synthesize ATP. As H^+ pass through this multisubunit protein, they cause transmembrane subunits to spin against subunits that bind ADP and P_i . The energy generated is used to create a chemical bond between ADP and P_i . It is estimated that one molecule of glucose produces, through glycolysis, the citric acid cycle, and oxidative phosphorylation, 30 ATP molecules [4].

DECODING THE GENOME (I): TRANSCRIPTION

The information necessary for the homeostasis and survival of a cell (and by extension multicellular organisms) and its progeny is encoded in DNA. Specific DNA sequences or **genes** direct the synthesis of RNA and proteins that carry out most of the cell's functions. DNA is a double-stranded helix composed of four different nucleotide monomers: adenine (A), cytosine (C), guanine (G), or thymine (T). Each nucleotide consists of a deoxyribose sugar containing a phosphate group at the 5' carbon, and a purine (A or G) or pyrimidine (C or T) base at the 1' carbon. The nucleotides are linked by phosphodiester bonds between the 5' hydroxyl group of one sugar to the 3' hydroxyl group of another. These sugar-phosphate linkages form the backbone of the DNA double helix. The nucleotide bases of the two strands are oriented inward from the backbone. Each purine base complementary pairs with a pyrimidine through hydrogen bonding; adenine pairs with thymine and guanine pairs with cytosine. Complementary base pairing only occurs if the two strands are antiparallel to each other so that one strand runs spatially in a 5' to 3' direction while the sister strand runs 3' to 5'. The helix conformation is primarily the result of intrabase pairing and interbase stacking.

The genome is very large, and in order for the relatively small nucleus to contain the entire genome, the DNA is packaged and condensed into **chromatin**. The DNA is wound approximately 1.7 turns around an octamer of **histone** proteins. This DNA-histone unit is called the **nucleosome**. These repeating nucleosomal units are further compacted and folded to several degrees, resulting in a tightly packaged **chromosome**. All processes involving DNA are governed by the structural state of chromatin. The structural modulation of the chromatin structure is a dynamic process known as **chromatin remodeling**, which involves nucleosome shifting and histone modification. Chromatin remodeling affects gene expression without altering the DNA sequence, and is therefore an **epigenetic** phenomenon.

DNA Replication

The genetic code must be heritable and theoretically error free. Dividing cells must maintain genomic integrity to ensure the survival of the daughter cells and subsequent cellular generations. The process by which a cell copies the genetic code is replication.

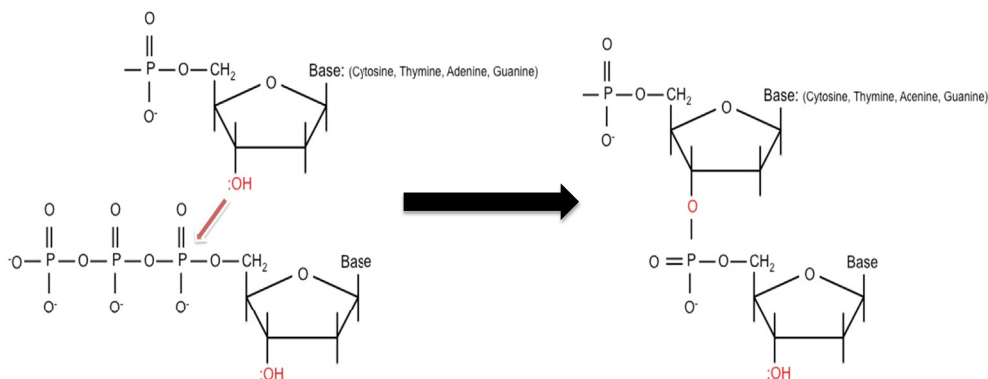


Figure 5 Nucleotide synthesis occurs in a 5' to 3' direction. 5' phosphate groups of nucleotides are nucleophilically attacked by the 3' hydroxyl group of the last nucleotide in the growing chain. Once added, the new nucleotide can attack the 5' phosphate group of a new nucleotide.

Double-stranded DNA is split into two “parent” or template strands. Each of the two strands is then complementarily replicated by replicative proteins. The end result is two double strands of DNA, each duplex containing one template strand and one copied strand. This is known as semiconservative replication.

Nucleotides are complementarily paired to the template strand in a 5' to 3' direction (Figure 5). That is, DNA polymerase links nucleotides through facilitating the nucleophilic attack of the 3' hydroxyl group on the phosphate adjacent to the 5' carbon of ribose. All nucleotide synthesis progresses 5' to 3'. So while replication occurs on the template strand (unwound in a 3' to 5' direction) or **leading strand** in a continuous 5' to 3' direction, how does replication occur on the lagging strand, which is unwound 5' to 3'? On this strand, the **lagging strand**, replication occurs semidiscontinuously in fragments known as **Okazaki fragments**. These fragments are ultimately linked by **DNA ligase**.

In eukaryotic organisms, the beginning site of replication is called the origin of replication (Figure 6). This area is bound by the origin recognition complex (ORC), which initiates replication. Replication activator protein (RAP) subsequently binds, along with replication licensing factors (RLF) which allows replication to proceed. This ORC–RAP–RLF (prereplication complex; preRC) complex is phosphorylated by cyclin-dependent kinases, resulting in the dissociation of RAP and RLF proteins (Figure 6).

The helicase protein unwinds the DNA, forming the replication fork. The replication sites are opened by two helicases unwinding in separate directions, forming a replication bubble, which replicate outward toward other growing replication bubbles (Figure 6). Single strand binding proteins localize to the unwound single strands, protecting against reannealing or nuclease activity. DNA polymerase is the primary enzyme which catalyzes the linking of the 3' hydroxyl group of the end nucleotide to the 5' phosphate of



Figure 6 Multiple origins of replication. The replicating genome features numerous replication bubbles replicating simultaneously. Multiple sites of replication ensure synthesis of the entire genome within an acceptable timeframe.

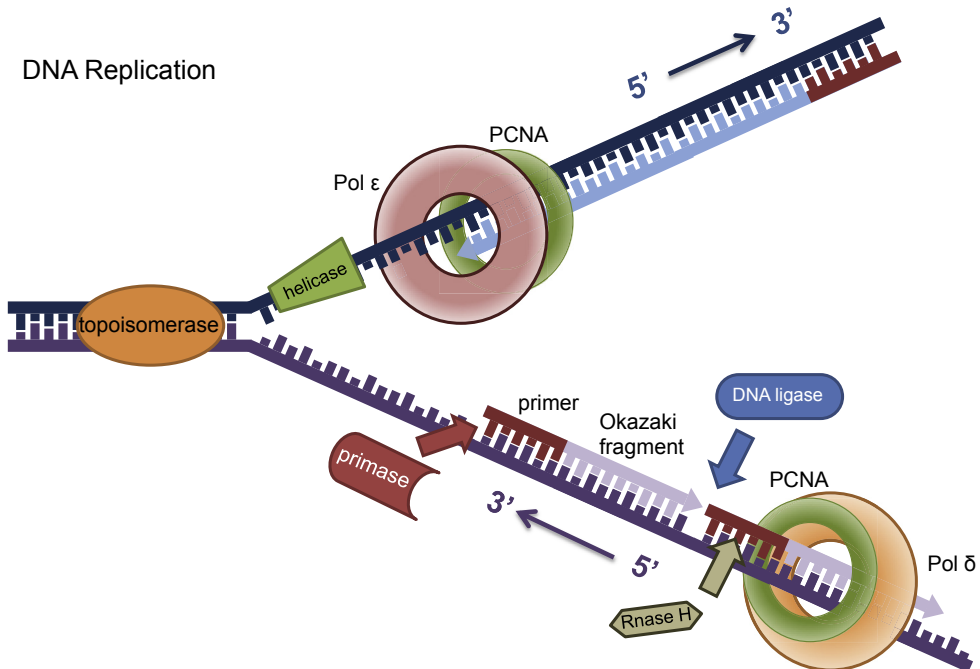


Figure 7 DNA replication. DNA replication is initiated by topoisomerase and helicase enzymes which unravel the DNA double strand. Primase synthesizes a short RNA primer on which DNA polymerase, aided by the PCNA stabilizer, adds nucleotides to the forming daughter strand. DNA polymerase synthesizes in a 5' to 3' direction. For the lagging strand, short regions of DNA are synthesized, called Okazaki fragments.

nucleotide to be added. This results in the formation of a phosphodiester bond between the nucleotides. There are several eukaryotic DNA polymerases. DNA polymerase III (Pol III) is a multisubunit polymerase which mediates nucleotide polymerization. Several of the subunits play various roles in replication: α , β , γ , δ , and ϵ . Pol III ϵ and δ are the polymerases primarily responsible for nucleotide strand synthesis. Pol III ϵ catalyzes the synthesis of the leading strand while Pol III δ synthesizes the Okazaki fragments of the lagging strand. Both polymers have polymerase (5' to 3') and exonuclease (3' to 5') activity (Figure 7). Pol III δ interacts with the proliferating cell nuclear antigen (PCNA), a homotrimeric protein that surrounds the DNA, functioning as a mobile clamp, positioning the

polymerase to the template strand during replication. Eukaryotic DNA cannot undergo replication without a short primer sequence in which to add nucleotides 5' to 3'. An RNA polymerase, primase, synthesizes this sequence, which is subsequently degraded by RNase H. The DNA double helix is unwound and DNA polymerase adds nucleotides complementary to the two parent strands to synthesize two complementary daughter strands.

The Cell Cycle

DNA replication is a carefully regulated process. In order to prevent replication from occurring at inopportune times, the cell restricts when DNA replication occurs. The process by which a cell duplicates the genome and divides into two daughter cells is known as the cell cycle. The cell cycle consists of two phases: interphase, during which the cell builds metabolic stores and duplicates the genome; and mitosis, the phase in which the duplicated and identical genomes are separated. The last act of mitosis is actual division of the cell into two identical daughter cells, called cytokinesis.

A newly-divided cell must “decide” whether to continue with cellular division, or exit the cycle into a nonproliferative resting phase or postmitotic differentiation. Mitogenic factors may induce a cell to enter into the mitotic pathway, while antimitogenic factors, or alternatively the absence of mitogenic factors, can cause the cell to withdraw from the cell cycle into the quiescent G_0 resting phase. The cell must prepare for mitosis by increasing in size, increasing the number of constituents of the cell, and replicating the genome. This preparatory stage of cell growth is known as the **interphase**. During the **first gap phase (G_1)**, the cell commits itself to mitosis. The **synthetic phase (S)** occurs next, in which the genome is duplicated. After DNA replication, the cell delays mitosis by entering into the **second gap phase (G_2)**, wherein the cell prepares for mitosis. These three phases are collectively known as interphase (Figure 8).

Lastly, the cell enters into **mitosis**, which is composed of six distinct stages leading to cellular division. In **prophase**, the chromosomes of the nucleus condense and become visible, and the centromeres assemble for spindle formation. In **prometaphase**, the duplicated chromosomes interact with the microtubule fibers of the spindle and the nuclear membrane dissolves. In **metaphase**, the chromosomes, attached to the spindle microtubules, align along a bisecting plane in the middle of the cell. The individual chromatids are pulled apart during **anaphase** to opposite ends of the cell. In **telophase**, the chromatids cluster and de-condense, around which a new nuclear membrane forms. The last stage is **cytokinesis**, where the cytoplasm divides into two new identical daughter cells (Figure 8).

1. **Interphase:** chromosomal duplication

- a. **G_1 (first gap) phase:** growth rate and metabolic activity increase to meet the energy demands of replication and mitosis
- b. **S (synthetic) phase:** DNA replication
- c. **G_2 (second gap) phase:** preparation for cell division

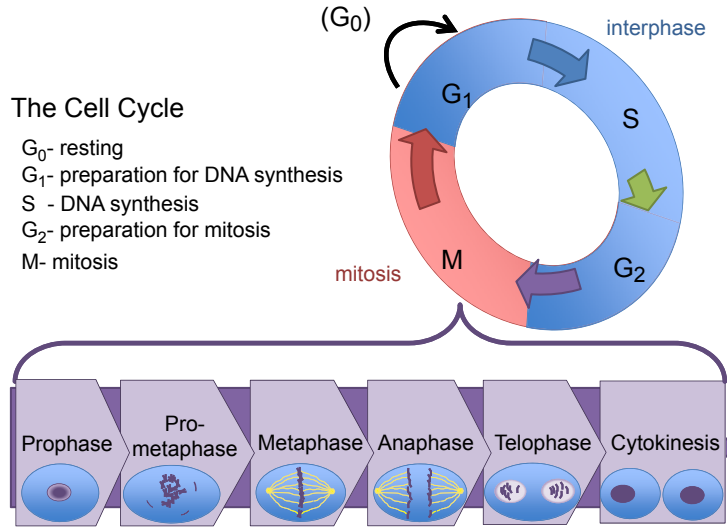


Figure 8 The phases of the cell cycle. The cell cycle is divided into two phases, “mitosis,” the cellular division phase, and “interphase,” where the genome is replicated and the cell prepares for division. Interphase consists of the G_1 , S , and G_2 stages. The cell commits to resting phase (G_0), or to cell division during the first “gap” phase, G_1 , and prepares for the metabolic demands of DNA replication. Synthesizing the genome occurs during S phase, which leads into the second gap phase, G_2 . G_2 phase prepares for cellular division. Mitosis is broken down into several phases: prophase, prometaphase, anaphase, telophase, ending in cellular division through cytokinesis.

2. Mitosis: nuclear and cellular division

- a. Prophase: chromosomal condensation and spindle formation
- b. Prometaphase: nuclear membrane disintegrates and chromosomes interact with spindle
- c. Metaphase: chromosomes align in the middle of the spindle
- d. Anaphase: chromatids separate and move to the opposite ends of the spindle
- e. Telophase: nuclear membranes form around the two chromosome groups
- f. Cytokinesis: the cell splits into two daughter cells

Regulation of the Cell Cycle

A multitude of signals that activate or inhibit cell proliferation and growth converge to regulate the cell cycle. The cell cycle is essential to cellular homeostasis and generational preservation of the genome, the regulatory burden of DNA replication is a heavy one, and aberrant DNA replication and DNA damage are implicated in many diseases. To ensure genomic fidelity, the stages of the cell cycle must proceed without error. The mechanism by which DNA replication is temporally regulated must be not only capable of controlling the rate of DNA replication, but must be able to halt and inhibit replication in the face of DNA damage. The cell cycle is a linear process, each phase dependent

upon the previous phase. Cell cycle checkpoints exist to ensure each phase is completed properly before progression to the next stage.

The cell controls cell cycle progression via **cyclin-dependent kinases (CDK)** that partner with temporally expressed enzymes called **cyclins**; the kinase activity of CDKs is increased when bound to regulatory cyclins (Figure 9). While CDKs are constitutively expressed, cyclin expression and degradation is regulated by phases of the cell cycle, ensuring phase-dependent expression of cyclins. Combinations of cyclins and CDKs confer target specificity. Progression through G_1 is managed by CDKs 4 and 6, which interact with cyclin D family members. Once the R-point is passed, members of the cyclin E family associate with CDK2 to usher in S phase. Progression of S phase is mediated by cyclin A, which binds to CDK2 in place of cyclin E.

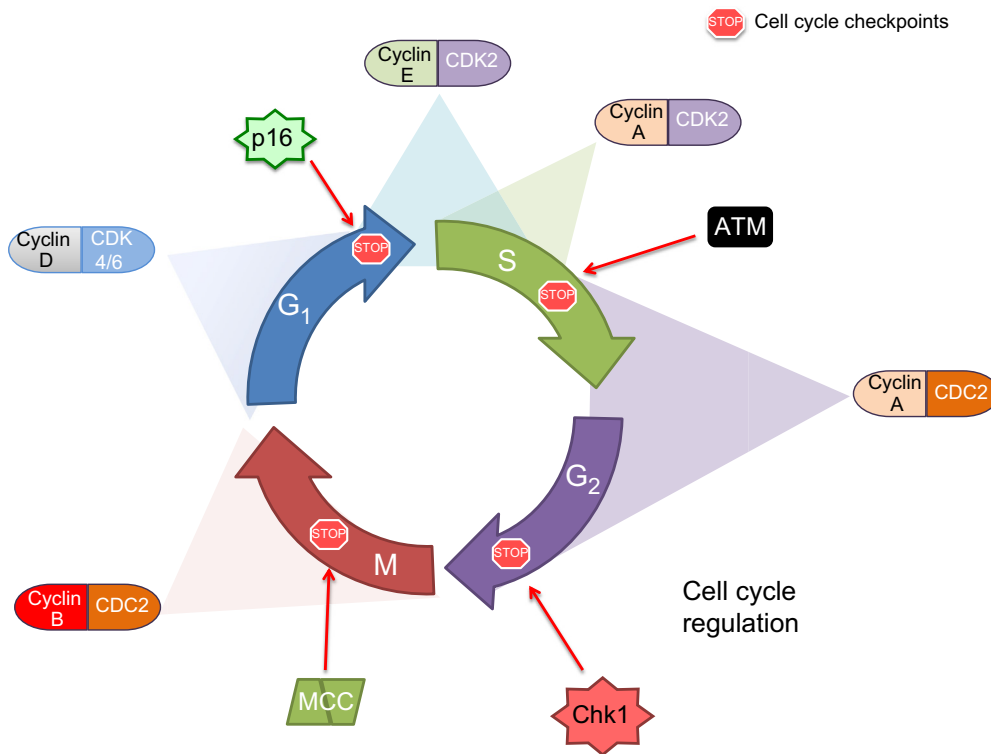


Figure 9 Regulation of the cell cycle. Progression through the stages of the cell cycle are facilitated by cyclin dependent-kinases (CDKs), which are activated by interaction with temporally-regulated cyclins. Each cyclin/CDK heterodimer regulates a different phase of the cell cycle. Cell cycle checkpoints exist in each stage to prevent premature progression of the cell cycle, thus ensuring that errors in the cell cycle or exogenous DNA damage are repaired before moving on to the next stage. Multiple enzymes are responsible for halting nuclear and cell division. Ataxia telangiectasia mutated (ATM), checkpoint kinase 1 (CHK1), mitotic checkpoint complex (MCC), and cyclin-dependent kinase inhibitor 2A (p16) are detectors and signaling mediators of DNA damage.

Near the end of S phase, cyclin A leaves CDK2 for CDK1, while next in G₂ phase CDK1 switches cyclin A with cyclin B, ultimately triggering mitotic events. The G₁ checkpoint, or restriction point, blocks entry into S phase if there is damage to the genome. The S checkpoint halts DNA replication if damage is detected, and if DNA synthesis is not completed, the cell is blocked from entering mitosis at the G₂ checkpoint. There also exists a checkpoint in mitosis phase which ensures proper segregation of the chromatids.

The G₁ phase is important in the cell cycle because it is during this window that the cell must commit to division, or withdraw. The G₁/S checkpoint, or **restriction point**, halts progress into S phase (Figure 9).

DNA Damage Response

Cells are continuously subjected to DNA damage from exogenous (environmental toxicants) and endogenous (oxidative stress and aberrant replication) sources. In order to maintain genomic integrity, the cell has evolved a **DNA damage response (DDR)** pathway, a multitier signaling pathway involving multiple, functionally diverse proteins. **Sensors** detect DNA lesions, **transducers** propagate the DDR signal, and **effectors** initiate the proper response, such as cell cycle checkpoint arrest, DNA repair, or **apoptosis** (Figure 10). It is essential for cell viability that these processes be prompt and efficient; repression or errors in DNA damage detection, signaling, and repair lead to genomic instability which is a major pathogenic process in diseases such as cancer. The multifactorial nature of the DDR pathway imbues the pathway with complexity that allows it to be finely tuned and controlled, but also leaves it susceptible to toxic alterations given the multitude of factors involved.

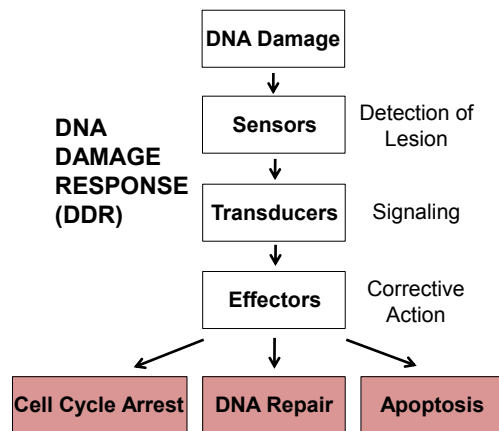


Figure 10 *DNA damage response (DDR) signaling pathway.* Sensors detect DNA damage and activate a cascade of signal transducers, which results in activation of DDR effectors which carry out the appropriate response, such as cell cycle arrest and DNA repair, or apoptosis.

DNA can be damaged in several ways. The strand itself may be severed, as in the case of **double strand breaks (DSB)**. Individual bases may be modified through oxidation, alkylation, methylation, hydrolysis, and deamination. Bases may also be modified by bulky adducts. The DDR must not only muster the correct repair process, but must identify which lesion has occurred in order to mount the proper repair pathway.

DNA Damage Detection and Response

The DDR pathway is responsible for the detection of deleterious physiochemical alterations to DNA and the mustering of a rapid, appropriate response. Cells respond to DNA damage in several ways, depending upon the type and severity of the alteration. The DNA repair system may be activated, or in some cases the cell may undergo programmed cell death, or apoptosis. The DDR is a cellular signaling pathway and most repair pathways utilize many of the same factors to detect DNA damage.

DNA Damage Repair

DNA damage repair is a multifactorial collection of processes which vary depending upon the type of damage involved. Lesions that spatially distort DNA such as adducts are repaired by **nucleotide excision repair**. In **base excision repair**, DNA glycosylase removes damaged bases, while the phosphodiester bond is cleaved by endonucleases, and the correct base is added by DNA polymerase. DNA ligase then seals the phosphodiester backbone. DNA DSB are the most cytotoxic and mutagenic DNA lesions. DSB are repaired by **homologous repair (HR)** or **nonhomologous end joining (NHEJ)** repair. NHEJ is more common, but is error prone. HR displays more fidelity, employing a homologous sequence as a template for repair. Mismatch repair (MMR) replaces mispaired bases that are damaged and if not replaced would result in heritable mutations. MMR is usually needed for errors in DNA replication and recombination. Chromatin remodeling and epigenetic alterations are essential to DNA damage repair, given that repair enzymes need access to sites of DNA damage. Histone modifications play a major role in DDR; the phosphorylation of histone H2AX is widely accepted as a marker of DNA damage.

The Mechanisms of Transcription

Genes, through expression of their protein products, regulate cellular homeostasis and responses to extracellular stimuli. A finely tuned mechanism is necessary for the intricate control of gene expression, given the vast diversity and number of gene-encoded proteins necessary for cellular survival at any given time. Evolution has provided an incredibly complex yet inventive set of mechanisms that control the process whereby the DNA sequence is “read” by specialized transcription factors and an RNA transcript is synthesized, which is subsequently translated into protein. The mechanisms which control

which particular gene undergoes transcription, and when, are multifactorial in nature. The enzymes that assemble the transcript from the DNA sequence are regulated by sequence motif **enhancer elements** within the sequence, allowing specificity in binding transcriptional activators and repressors. This mechanism works in tandem, or is regulated by epigenetic alterations such as histone modifications, DNA methylation, and miRNAs. The interdependence of this transcriptional network allows for extreme selectivity and sensitivity in regulating gene expression.

The Gene Landscape

Genes are divided into specific regions that play major roles in the transcriptional process (Figure 11). The **core promoter** is the region where transcription is initiated. It contains the **transcription start site (TSS)** (+1 bp) which is the first nucleotide transcribed into RNA, and several sequence elements that regulate recruitment and positioning of the transcriptional machinery. The presence of each element varies from gene to gene depending upon the frequency at which a gene is activated; the **TATA box** (−28 to −34 bp) is found on many frequently activated genes. Other sequence elements in the core promoter include the transcription factor IIB (TFIIB) **recognition element (BRE)**, the **initiator element (Inr)**, and **downstream promoter element (DPE)** (+28 to +34 bp). The Inr element surrounds the TSS, though this sequence is not well-conserved. These elements aid in recognition of the TSS. Upstream of the core promoter is the **proximal promoter region** (−250 bp) that regulates transcription through the binding of **transcription factors (TF)**. Scattered far upstream and downstream of the core promoter are distal enhancer and silencing elements that also bind transcription factors and promote either activation or repression of transcription. Downstream of the TSS is the gene body, or coding region. This region contains the sequence that is transcribed into RNA and is composed of **introns**, regions removed during RNA splicing, and **exons** that are spliced together to form the mature mRNA.

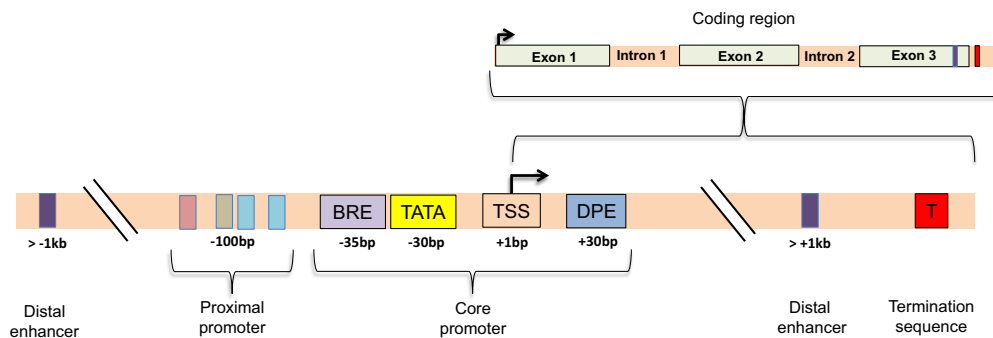


Figure 11 The gene landscape. A gene is composed of several regions which vary in function. The coding region contains the genetic information that encodes for proteins.

Enhancers, Silencers, and Transcription Factors

Enhancer and silencer elements regulate transcription through the binding of a multitude of TFs that activate or repress transcription. These transcription factor-binding elements influence transcription irrespective of location; they may lie proximal to the core promoter or several hundreds of base pairs from the TSS, or even on other chromosomes. Given the limited number of transcription factors compared with the number of genes under transcriptional regulation, and the repetitious use of the same transcriptional machinery, it is the enhancer and silencing motifs found proximally and distally that provide much of the temporal and spatial regulation of transcription. Through the binding of transcription factors, enhancer elements direct which genes are to be transcribed, when transcription takes place, for how long, and at what level of intensity.

The sequence motif of enhancer elements determines the specificity of transcription factor binding, though these sequences are usually degenerate. Variants of consensus sequences may dictate the strength of association with a particular transcription factor, or preferential binding of specific dimerization partners. Sequences may also alter the conformation of a bound transcription factor, affecting its activity. Transcriptional **activators** may promote preinitiation complex (PIC) formation, regulate primary transcriptional events such as initiation and elongation, and recruit chromatin modifiers. **Coactivators** have similar functions, but do not bind directly to enhancer elements, instead associating with enhancer-bound activators.

Transcription factors aid in the recruitment of the RNAPII transcriptional complex and recruit chromatin remodelers. How do distal enhancers regulate transcriptional activity at the core promoter, several hundred base pairs away? The multisubunit coactivator Mediator complex binds to distal transcription factors and causes the DNA to loop so that Mediator is in proximity to RNAPII, to which it binds at the C-terminal domain (CTD). In addition to bridging the distance between distal enhancers and the TSS, Mediator also mediates the phosphorylation of RNAPII at serine 5 (Figure 12).

Preinitiation Complex (PIC)

The **PIC** is composed of several subunits, termed **general transcription factors (GTF)**. These units are the first to assemble at the core promoter. They clear the way for RNA polymerase II (RNAPII) binding by specifying the location of transcription, the orientation of the transcriptional complex, and serve as a platform to which RNAPII locates and binds. They also aid in opening the DNA double strand into an open complex. While bacteria employ only one GTF, RNAPII-mediated transcription in eukaryotes involves several:

1. **TFIID** is composed of the **TATA-binding protein (TBP)** and several TBP-associated factors (TAF). TAFs recognize core promoter elements while TBP binds to the TATA box or in the case of TATAless promoters, the DPE or Inr. The binding of TBP to the TATA box partially unwinds and bends the DNA at the core promoter region.
2. **TFIIA** associates with TBP, stabilizing the TBP–DNA interaction.

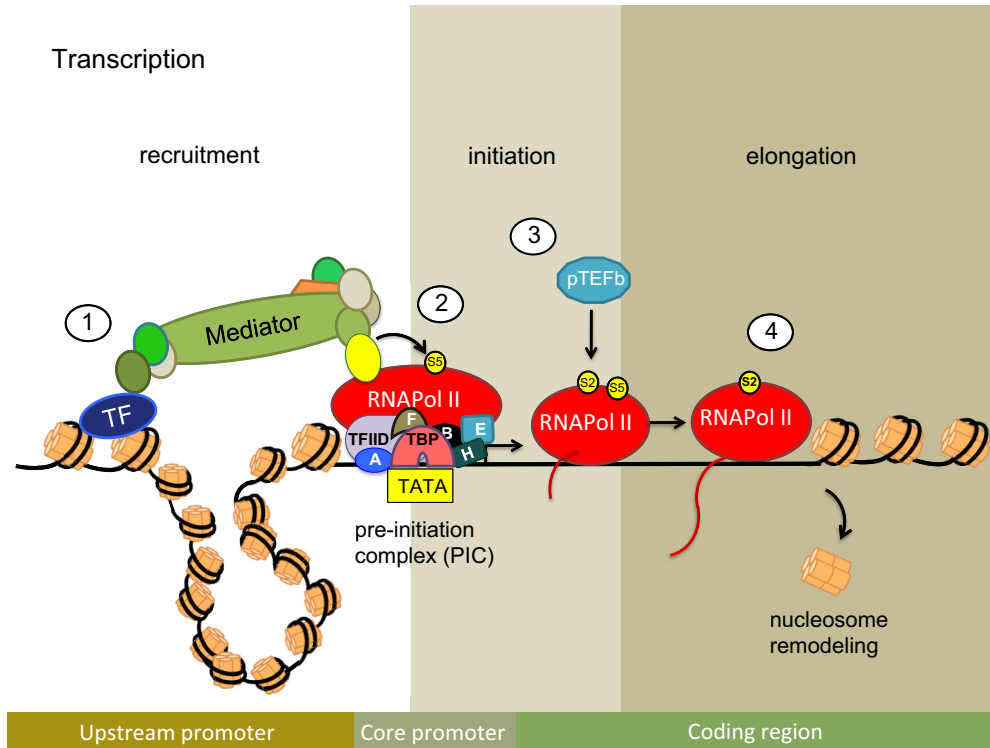


Figure 12 Transcription.

3. **TFIIB** binds the BRE and recognizes the TSS, and links RNAPII with TFIID by binding to TBP and the recruited RNAPII.
4. **TFIIE** is involved in the melting of promoter DNA into an open complex.
5. **TFIIF** binds to RNAPII, stabilizing RNAPII within the PIC.
6. **TFIIH** subunits induce formation of the transcription bubble by promoter melting, in addition to phosphorylating RNAPII at the CTD at serine 5, inducing promoter clearance.

RNA Polymerase

Both prokaryotic and eukaryotic cells employ the multisubunit **RNA polymerase (RNAP)** complex as the main effector of transcription. Eukaryotic cells employ three types of RNAP (I, II, and III) for synthesizing three different classes of RNA. RNAPII, that synthesizes mRNA, is composed of 10–12 subunits. The RNAPII holoenzyme contains the polymerase associated with GTFs. Functionally, RNAPII is composed of four multisubunit mobile elements:

1. The **core** contains the cleft that is surrounded by the other units; the cleft holds the incoming single DNA strand at the active site.

2. The **clamp** is a mobile unit connected to the core that opens and closes access to the cleft, stabilizing the DNA–RNA hybrid with the polymerase.
3. The **shelf**
4. The **jaw lobe** grips the DNA downstream of the active site.

RNAPII passes the template strand through the core cleft, adding base pairs complementary to the template strand to form an RNA strand which is an exact copy of the coding strand, except that RNA contains uracil in lieu of thymine. RNAPII transcribes 3 to 5' along the template strand, producing a 5 to 3' RNA strand. Several RNAPIIs can transcribe a single gene, increasing the amount of RNA produced. The transcription bubble is the unwound DNA structure that allows RNAPII to transcribe along the single template strand. As RNAPII progresses, the DNA downstream is unwound. Subunits of TFIIH have helicase and ATPase properties which facilitate bubble formation. Other functional elements of importance are the **C-terminal domain (CTD)**, which is a repeating heptad sequence that is phosphorylated at specific residues (serine 2 and 5), which govern the initiation and elongation steps. Phosphorylation of this scaffold region regulates transcription by releasing inhibitory factors.

Briefly, GTFs bind to a core promoter sequence (TATA), located upstream of the TSS. This closed, multiunit PIC recruits and binds RNAPII. A small section of the DNA double strand separates, and the template strand lies in the active site of the RNA polymerase. Initial attempts at RNA synthesis may be unsuccessful due to the low stability of the short DNA–RNA strand in the active site; once RNAPII synthesizes an RNA strand which is stable, it leaves behind the PIC, “clearing” the promoter. RNAPII proceeds to synthesize.

This event initiates RNA synthesis where the RNAPII complex is released from the promoter region. After this first step of recruitment and initiation, RNAPII undergoes phosphorylation at serine 2 in the CTD (RNAPIIS2), resulting in productive transcriptional elongation along the coding region. Transcription ends with a termination sequence motif (AAUAAA) that halts RNAPII. RNAPII is subsequently dephosphorylated and recycled into a new transcriptional cycle.

1. **Chromatin remodeling.** Transcriptional activators bind and induce chromatin remodeling at the promoter region to allow transcriptional machinery access to the transcription start site. Promoters of genes undergoing continual transcription are usually nucleosome-free.
2. **Preinitiation complex assembly.** GTFs assemble on the core promoter and recruits RNAPII.
 - a. TFIID binds to the core promoter through the TBP subunit; binding may occur at specific elements such as the TATA or Inr. In promoters lacking specific elements, TAFs associate with the promoter and induce nonspecific binding of TBP. TBP binding to the TATA opens the minor groove, bending the DNA at an 80° angle.
 - b. TFIIA binds to TBP, stabilizing the interaction between TFIID and DNA.

- c. TFIIB binds to the core promoter through BRE and positions the DNA for entry into the active site. This TFIID/A/B complex recruits RNAPII and TFIIF simultaneously.
 - d. TFIIE joins the complex and recruits TFIIH, aiding in opening the DNA double strand, called promoter melting. TFIIF induces torsional strain, forming the transcription bubble.
 - e. TFIIH contains a helicase subunit that unwinds the DNA into two strands; it binds to the template strand to ensure transcription of the correct strand. The catalytic activity of TFIIH is dependent upon TFIIE activity, which it interacts with. TFIIH promotes formation of the transcriptional bubble. TFIIH also contains a kinase subunit that phosphorylates RNAPII at the CTD to initiate transcription.
 - f. The bridging protein mediator links distal transcriptional activators with the RNAPII complex.
3. **Initiation and clearance.** Phosphorylation of the CTD at serine 5 (RNAPIIS5) by TFIIH or a mediator subunit induces RNAPII to clear the PIC and begin transcribing RNA from the DNA template strand; RNAPII may pause approximately 50 bp downstream.
4. **Elongation.** Paused RNAPII is phosphorylated at serine 2 (RNAPIIS2) on the CTD by the positive transcription elongation factor b, causing dissociation of the negative elongation factor, allowing RNAPII to proceed into continuous, productive elongation. Chromatin remodeling of nucleosomes occurs as RNAPII transcribes through the coding region.
5. **Termination.** Upon reaching the termination signal, the transcribed RNA is released and RNAPII is dephosphorylated, dissociating from the gene to be recycled for subsequent cycles of transcription.

DECODING THE GENOME (II): TRANSLATION

The process of deciphering the DNA code into proteins, which will carry out the myriad functions of the cell, is a multistep process. Transcription “reads” the DNA, producing an RNA “blueprint.” This RNA blueprint is then used to assemble a polypeptide chain. This chain of amino acids will then undergo folding and further processing, producing an active protein, whether catalytically active enzyme, or a structural component like actin. Compared with transcription, the process of translation is peculiar in that the RNA is processed by a molecular machine called the ribosome, which is composed of specialized ribosomal RNAs. Other nonprotein coding RNAs, aptly termed transfer RNA, are used to transport the individual amino acids to the ribosome where they are attached to the growing polypeptide chain.

RNA

If an apt metaphor for DNA is a set of “master plans” which describe how cellular functions are to be executed through the generation of proteins, then RNA is the actual “blueprint” which directs the detailed assembly of proteins. However, RNA is not simply a schematic for protein assembly, but serves as effectors of cellular function as well.

1. **Messenger RNA (mRNA)**. Contains the amino acid sequence for the assembly of proteins.
2. **Noncoding RNA (ncRNA)**. RNA that does not code for protein assembly.
 - a. **Ribosomal RNA (rRNA)**. The structural and catalytic RNA component of ribosomes which direct protein synthesis.
 - b. **Transfer RNA (tRNA)**. Structural RNA that transports amino acids to the ribosome for incorporation into the peptide chain.
 - c. **Regulatory RNAs**. RNA transcribed from DNA that is not utilized for protein synthesis but instead regulates gene expression, and protein translation.
 - Micro RNA (miRNA)
 - Small interfering RNA (siRNA)
 - Enhancer RNA

Processing of mRNA

In eukaryotes, after the pre-mRNA has been transcribed, extensive modifications take place. The 5′ end undergoes **capping**, the 3′ end undergoes **polyadenylation**, and the coding region undergoes **splicing**. Furthermore, tRNA and rRNA undergo posttranscriptional modifications; however, for the scope of this chapter only mRNA processing will be described.

Capping

The 5′ end is capped with a guanylate residue methylated at the N-7 position. This residue is attached to the adjacent residue by a 5′ to 5′ triphosphate linkage, and this adjacent residue may be methylated as well. Capping prevents exonuclease degradation.

Polyadenylation

Polyadenylation of the 3′ end occurs before the mRNA leaves the nucleus. This polyadenylate tail, around 100–200 nucleotides long, protects the mRNA from the degradatory action of phosphatases and nucleases. While prokaryote genes are continuous, that is, each base pair in the DNA sequence is conserved in the mRNA, eukaryotic mRNA is smaller than the template DNA sequence due to the splicing out of introns and the fusing together of exons. The 5′ capping and 3′ polyadenylation are not translated. Export of mRNA from the nucleus into the cytosol relies on polyadenylation of the 3′ end of the strand.

Splicing

Splicing occurs in the nucleus. During transcription, the RNA associates with various proteins, forming ribonucleoprotein particles. Splicing cleaves the 5' and 3' ends of the intron sequences at splice sites, which are GU (5') and AG (3'). Another site involved in intron cleavage is known as the branch point, usually located 20–40 bp upstream from the 3' end. The branch sequence is variable, PyNPyPuAPy, except for the A. The process of intron removal first involves the looping of the 5' G looping downstream into contact with the invariable A of the branch site. The 2' hydroxyl residue of the A performs a nucleophilic attack on the phosphodiester backbone of the 5' G site. This causes the release of the upstream exon from the 5' end of the intron. The resultant AG site of the released 3' end of the exon then nucleophilically attacks the phosphodiester backbone of the 3' G site of the intron, thus fusing the two exons and releasing the looped intron. This process is mediated by **small nuclear ribonucleoproteins (snRNPs)**, composed of 100–200 nucleotide RNAs and ≥ 10 proteins, that bind to the pre-mRNA through complementary sequences at the splice and branch sites. Splicing is carried out by 50–60S particle called the **spliceosome**. Gene expression is regulated at several levels, transcriptionally, translationally, and even at the splicing stage. **Alternative splicing** of some genes gives rise to several different mRNA **isoforms**, which are translated into proteins of differing size and function.

Protein Synthesis

After processing, the mRNA is exported out of the nucleus to ribosomes, the primary site of protein synthesis where it is “read” and translated into protein. The mRNA is bound by the ribosome, and amino acids attached to tRNAs bind to the ribosome and mRNA, based on complementary pairing of the tRNA codons with the mRNA codons. The amino acids are linked together by peptide bonds, thus forming a growing polypeptide chain.

Amino Acids and Peptide Bonding

Amino acids are composed of a central α -carbon, which is bonded to a hydrogen atom, an amino group ($^+\text{NH}_3$), a carboxyl group (COO^-), and a distinct side chain (R) that confers specificity to each amino acid. The side chain groups are biologically important, and are classified by whether they are polar or nonpolar, and whether they contain basic or acidic residues. Amino acids are linked by covalent bonding between the α -carbon of an amino acid and the amino group of another.

Codons

The information necessary for protein assembly are encoded in mRNA through **codons**, a three base sequence. Specifically, a codon is a triplet nucleotide sequence. There are 64 possible codons, 61 of them representing the 20 amino acids, and the remaining three

serving as stop codons, which are translation termination sequences. While an amino acid may be represented by more than one codon, no single codon encodes more than one amino acid. Methionine and tryptophan are the only amino acids represented by a single codon. Some amino acids are represented by up to six codons. The first and second bases of the codon are usually conserved, with the third base variable, known as the “wobble” base. This variability prevents deleterious phenotypes caused by mutations of the third base of the codon. Codons base pair with the **anticodon** of tRNA. In addition to coding for methionine, AUG is the initiating codon.

Transfer RNA (tRNA)

Transfer RNAs (tRNAs) contain an anticodon sequence which pairs with complementary mRNA codons. The conserved, three-dimensional shape of tRNAs arises in part from the interaction between bases in different regions of the RNA sequence. These interactions give rise to the secondary “cloverleaf” structure of tRNA; this base pairing results in three stem-loops and an open-ended stem formed by the pairing of the 5′ and 3′ ends. The latter stem is known as the acceptor stem, which binds an amino acid. Of the three stem loops, the anticodon loop contains the three nucleotide base sequence which pairs with the mRNA codon during translation. There are more possible codons (64) than individual tRNAs. As mentioned previously, the third base of the codon (5′ to 3′) is the wobble base which allows for an amino acid to be coded by several different codons. The first base of the anticodon (5′ to 3′) is a wobble base; if the base is G, U, or I (inosine), there exists variations in hydrogen bonding which allows the anticodon to base pair with more than one codon. While several amino acid types may bind to a tRNA, only one amino acid at a time may bind. Amino acids are attached to tRNA at the acceptor stem by the enzyme aminoacyl-tRNA synthetase in two steps. This enzyme catalyzes the hydrolysis of ATP, covalently bonding an amino acid to AMP, forming aminoacyl-AMP which remains bound to the enzyme. This aminoacyl residue is subsequently transferred and bound to the 3′ end of the acceptor stem through an ester linkage, forming aminoacyl-tRNA. Specificity is achieved by there being a synthetase for each type of amino acid which recognizes the tRNAs specific for that amino acid. However, a synthetase may recognize a structurally similar amino acid; this possible error is corrected at the second step. If the incorrect amino acid is attached to a tRNA, the synthetase detects this error through an editing site and hydrolyzes the incorrect amino acid. Another level of specificity is achieved by the synthetase being able to recognize specific tRNAs.

Ribosomal RNA (rRNA) and Protein Synthesis

The ribosome is the site of protein synthesis. The ribosome is composed primarily of two ribosomal RNA (rRNA) subunits, also known as ribozymes, catalytically active RNA. The smaller subunit reads the mRNA transcript while the larger subunit facilitates the synthesis of the amino acid chain. There are three binding sites in the ribosome;

the aminoacyl (A) site, the peptidyl (P) site, and the exit (E) site. These sites are oriented in the context of the mRNA strand 5' to 3', E–P–A, due to the movement of the ribosome toward the 3' end of mRNA. The ribosome reads the mRNA by codon, that is, the ribosome moves along the mRNA three bases at a time.

Initiation of translation in eukaryotes is mediated by **eukaryotic initiation factors (eIF)**. The first step is the assembly of a 43S preinitiation complex. Methionine (Met) is the first amino acid of the polypeptide chain, due to the initiation codon being AUG, methionine is transferred by tRNA_i, an initiator tRNA. Met-tRNA_i, in a complex with eIF2 and GTP, is transferred to the 40S ribosomal subunit, which is bound by eIF1A and eIF3. The second step in initiation is the recruitment of mRNA to the ribosome. The 5' cap of the mRNA orients the ribosome to an AUG codon. Binding of eIF4E to the 5' cap brings the cap into proximity of the poly A tail by binding to the poly A binding protein (Pab1p). This bridge between the poly A tail and the 5' cap is also mediated by the association of several other eIFs with the eIF4E–Pab1p complex. The 40S–eIF complex, upstream of the start codon, moves 5' to 3' along the mRNA strand until it encounters the correct start codon, which is identified by two determining factors: the presence of the Kozak sequence, a consensus sequence surrounding a start codon (ACCAUGG), and the lack of a hairpin loop downstream of an AUG codon. In step three, the 80S initiation complex is formed by association of the 60S ribosome with the 48S preinitiation complex, which consists of the mRNA–eIF complex. The GTP recruited by Met-tRNA_i in step one is hydrolyzed, releasing the initiation factors (Figure 13).

Translation occurs in three steps similar to transcription. During the **initiation phase of translation**, the mRNA is the focus of a ribosome assembly, and the first tRNA is attached to the start codon. The second phase of **elongation**, the tRNA transfers an amino acid to the tRNA of the next codon. The ribosome then proceeds to the next codon of the mRNA, assembling the amino acid chain. The **termination** phase sees the ribosome releasing the polypeptide upon encountering the stop codon.

Translation in eukaryotes occurs mainly on the rough ER. The aminoacyl tRNA synthetase is a multidomain enzyme which catalyzes the ATP-dependent bonding of an amino acid to the corresponding tRNA for transport to the ribosome, called an aminoacyl–tRNA. Once at the ribosome, the aminoacyl–tRNA is complementary base paired to the corresponding mRNA codon. The aminoacyl site binds the aminoacyl–tRNA with the complementary mRNA codon (Figure 13, 2). The peptidyl site stabilizes the aminoacyl–tRNA with the enlarging polypeptide chain. A peptide bond is made between the amino acid of the tRNA at the aminoacyl site and the amino acid at the peptidyl site (Figure 13, 3). The exit site holds the tRNA after transfer of the amino acid. The tRNA at the peptidyl site, now without an amino acid, is transferred to the exit site. The tRNA at the aminoacyl site is transferred to the peptidyl site (Figure 13, 4). The tRNA at the exit site is ejected (Figure 13, 5) and another aminoacyl–tRNA enters the aminoacyl site (Figure 13, 1) to repeat the process.

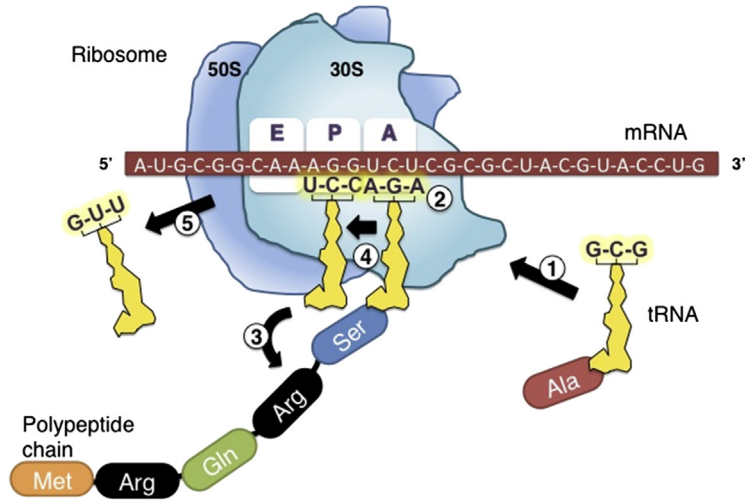


Figure 13 Translation. Aminoacyl-tRNA transports amino acids to ribosomes (1). Aminoacyl-tRNA anticodon pairs with mRNA codon and binds to the aminoacyl site (A) (2). Through peptide bond formation, polypeptide chain of tRNA at peptidyl site (P) is transferred to amino acid of tRNA at aminoacyl site (3). tRNA containing polypeptide chain transfers from the aminoacyl site to the peptidyl site, while the exit site (E) tRNA without an amino acid is released from the ribosome (4). New aminoacyl-tRNA bearing the next coding amino acid is transported to the ribosome for binding at the open aminoacyl site (5).

Protein Structure and Folding

The function of a protein is dependent upon its three-dimensional structure. The polypeptide chain synthesized by the ribosome undergoes folding to achieve its final conformation. Misfolded proteins may be functionally inert, or dysfunctional, and susceptible to degradation; several pathological states arise from protein misfolding [5]. Protein folding at essence is the attainment of a thermodynamically favorable arrangement of the amino acid residues. The structure of a protein can be categorized into four different levels. The **primary** structure is the specific amino acid sequence of the polypeptide chain. The **secondary** structure is the spatial arrangement of the peptide backbone, stabilized by hydrogen bonding between amide and carbonyl groups of the peptide chain. The α -helix and β -pleated sheets are common secondary structures. **Tertiary** structure is the three-dimensional conformation, taking into account the amino acid side chains. The arrangement of peptide subunits to form a large multisubunit protein is known as the **quaternary** structure [6]. Protein folding may occur as the peptide chain is being synthesized at the ribosome, or after transport to the cytoplasm or in the ER [5].

CELLULAR HOMEOSTASIS (II): CELL DIFFERENTIATION, DEATH

Cells serve as the structural and functional units of tissues. Therefore, the fate of individual cells governs the homeostasis of tissues, and by extension, organ systems and the organism. Specialized cellular function depends upon the transformation, or

differentiation, of progenitor cells into specialized cell types. Cell division and death play central roles in tissue homeostasis, maintaining tissue mass and function. Proper development and survival depends upon the correct balance between cell division and death.

Differentiation

The physical characteristics and functional roles of a particular tissue are dependent upon the cell types that make up the tissue. The liver, for example, is comprised primarily of specialized hepatocytes that are structurally and functionally distinct from neuronal cells of the brain. How do stem cells transform into specialized cell types? What governs the process by which a stem cell transforms into a hepatocyte, but not a neuronal cell? The cellular process wherein a progenitor cell transforms into a different cell type is known as **differentiation**. Differentiation is mainly a transcriptionally regulated process. That is, the expression profile of a cell—which specific genes will be expressed and which will not—determines whether the cell will differentiate, and into what type of cell. Gene expression profiles are regulated by signal transduction pathways; the PI3K [7] and transforming growth factor- β [8] signaling pathways are mediators of cellular differentiation. At the transcriptional level, epigenetic alterations such as DNA methylation [9], histone modification [10], and miRNAs [11] also regulate differentiation.

Senescence

Damaged cells may undergo programmed cell death, or apoptosis; however, they may also enter a nonproliferative state known as senescence. Cellular stress, damage, and prooncogenic signaling may trigger senescence. In such cases, senescence serves to inhibit cancer progression, regulates wound healing, and limits pathological fibrosis. However, senescence may also play a role in development [12].

Apoptosis

Apoptosis, or programmed cell death, is covered in greater detail in Chapter 8 of this book. Briefly, apoptosis is the cell-initiated, controlled death of a cell regulated by mitochondrial dysfunction or signal transduction, initiating a series of protein–protein interactions leading to cell death [13]. Unlike necrosis, apoptosis is an ordered, stepwise series of events wherein the dissolution of the cell is contained so as not to damage neighboring cells.

Necrosis

Traumatic damage to the cell results in necrosis wherein the unwarranted dissolution of the cell membrane results in the release of cellular constituents into the extracellular matrix, resulting in inflammation and neighboring cell death. However, it has been shown that necrosis can be regulated, and that there are subclasses of necrosis [14,15].

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